Status	Finished
Started	Thursday, 3 September 2026, 3:11 PM
Completed	Thursday, 3 September 2026, 3:27 PM
Duration	15 mins 57 secs
Grade	**23.00** out of 25.00 (**92**%)

Question 1

Complete

Mark 2.00 out of 2.00

Consider an FSK signal at 10 Kbps in half-duplex transmission mode, the carriers are separated by 4 KHz. Fill in the blanks.

- The baud rate = 10 KBaud
- The minimum bandwidth = 14 KHz

Hint: Slide 42

FSK half duplex: N (bit rate) = S (baud rate)

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BW = (fc1-fc0) + Nbaud = 4KHz + 10 KBaud = 14 KHz

Question 2

Complete

Mark 3.00 out of 3.00

Consider an FSK in full-duplex transmission mode if the bandwidth of the medium 10 KHz , the upper carrier frequency is 20 KHz, the bit rate (N) is 2 Kbps. Fill in the blanks.

- The bandwidth for each direction is 5 KHz
- The baud rate (S) = 2 KHz
- As the Bandwidth (BW) = Baud Rate + (fc1-fc0), therefore the lower carrier frequency = 17 KHz

Bandwidth for each direction = 10KHz/2 = 5KHz

baud rate (S) = the bit rate (N)

fc0 = Baud Rate + fc1 - BW = 2 KHz + 20 KHz - 5 Kz = 17 KHz

Question 3

Complete

Mark 8.00 out of 8.00

Consider an FSK, an available bandwidth spanning from 1000 KHz to 3000 KHz. Fill in the blanks.

Hint: Slide 47

- The bandwidth = 2000 KHz
- The midpoint of bandwidth = 2000 KHz
- The lower carrier frequency (fc0) = 1500 KHz
- The upper carrier frequency (fc1) = 2500 KHz
- The upper and lower carriers are separated by fc1 - fc0 = 1000 KHz
- Define d = 1 for BW = (1 + d) S + 2*(Δf), while 2*(Δf) = fc1-fc0 = 1000 KHz
- Therefore: S = 500 KBaud
- The data rate = 500 Kbps

Bandwidth = 2000 KHz

BW = (1 + d) S + 2*(Δf) --> 2000 = (1 + 1) * S + 1000, S= 500 KBaud

Because baud rate (S) = the bit rate (N), N = 500 Kbps

Question 4

Complete

Mark 2.00 out of 4.00

What is the required bandwidth (Hz) for the following cases if we need to send 5000 bps?

Let d = 1.

- ASK: Bandwidth = 10000 Hz.
- FSK with 2Δf = 4KHz: Bandwidth = 14000 Hz
- QPSK: Bandwidth = 5000 Hz
- 16-QAM: Bandwitdh = 2500 Hz

Hint
ASK: BW = (1 + d) * S
FSK: BW = (1 + d) * S + 2Δf
QPSK: r=2--> S = N/r
16-QAM: r = 4 --> S = N/4

- ASK: S = N = 6000 Bauds
- ASK: BW = (1 + d) * S = 2 * 5000 = 10000 Hz

- FSK: S = N = 6000 Bauds
- FSK: BW = (1 + d) * S + 2Δf = 2 * 5000 + 4000 = 14000 Hz

- BPSK: S = N, BW = (1+d)S
- QPSK: r=2--> S = N/r = 5000/2 = 2500 Hz

Question 5

Complete

Mark 4.00 out of 4.00

What is the number of bits per baud for the following techniques?
Hint: find r = log2(L)

- ASK with 8 different amplitudes: 3 bits
- FSK with 4 different frequencies: 2 bits
- PSK with 16 different phases: 4 bits
- QAM with a constellation of 256 points: 8 bits

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Question 6

Complete

Mark 4.00 out of 4.00

Calculate the data rate (N) in Kbps for the given baud rate (S) and type of modulation
Hint: N = r * S

- 2000 baud/s, FSK

Answer: Data rate = 2 Kbps

- 2000 baud/s, ASK

Answer: Data rate = 2 Kbps

- 2000 baud/s, 4-ASK

Answer: Data rate = 4 Kbps

- 2000 baud/s, 16-QAM

Answer: Data rate = 8 Kbps